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Following Paper ID and Roll No. to be filled in your Answer Book.											
PAPER ID : 49906	Roll No.									0	8

B. Tech. Examination, 2024-25

(Even Semester)

ENGINEERING PHYSICS

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $8\times1=8$

   (a) Two independent sources could not produce interference, why?

   (b) What do you mean by diffraction of light?

   (c) Define specific rotation.

   (d) What are eigen values and eigen functions?

   (e) What are the outcomes of Laue's experiment?

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(f) What is numerical aperture of an optical fibre?

(g) What is displacement current?

(h) Define mass less particle.

SECTION – B

2. Attempt any two parts of the following : $2\times6=12$

(a) A thin film of soap solution is illuminated by white light at an angle of incidence $i=\sin^{-1}\left(4/5\right)$. In reflected light, two dark consecutive overlapping fringes are observed corresponding to wavelengths 6.1×10^{-7} m and 6.0×10^{-7} m. The refractive index of soap solution is 4/3. Calculate the thickness of film.

(b) Calculate the velocity and kinetic energy of a neutron having De-Broglie wavelength 1 Å.

(c) Find the mass and speed of 2 MeV electron.

(d) If earth receives 2 cal min^{-1} cm^{-2} solar energy. What are the amplitudes of electric and magnetic fields of radiation?

SECTION – C

Note :– Attempt all questions. Attempt any two parts from each questions. $8\times5=40$

3. (a) Discuss the formation of Newton's ring in reflected light. Prove that in reflected light the diameter of dark ring is proportional to the square root of natural number.

(b) Define resolving power. Derive an expression for the resolving power of grating.

(c) Describe the construction and working of NiCol prism.

4. (a) Derive time independent Schrodinger wave equation.

(b) What is Bragg's law? Describe Bragg's spectrometer?

(c) A particle is in motion along a line between x = 0 and x = a with zero potential energy. At points for which $x < 0$ and $x > a$, the potential energy is infinite. The wavefunction for the particle in the n^{th} state is given by $\psi_n = A\sin\frac{n\pi x}{a}$. Find the expression for the normalised wave function.

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5. (a) Show that velocity of plane electromagnetic wave in free space is given by $C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$.

(b) Explain the basic principle of optical fibre. Discuss fibre classification.

(c) Derive Maxwell's equations. Explain the physical significance of each equation.

6. (a) Show that the relative invariance of the law of conservation of momentum leads to the concept of variation of mass with velocity.

(b) Deduce the Lorentz transformation equations.

(c) Show that moving clock appears to be slowed down to the stationary observer.

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